

ABES Engineering College, Ghaziabad



Estd. 2000

B. Tech. First Year Curriculum

“Design Thinking, Innovation and Entrepreneurship”

Unit-5

IPR and Entrepreneurship

Syllabus

Unit 5: IPR and Entrepreneurship

Introduction to IPR, Types of IPR (copyright, patent, trademark, GI tag), advantages and disadvantages of IPR, prior art research techniques using tool such as Google Patent and other

Entrepreneurship concept, myths versus reality, entrepreneurship competencies, sensing business opportunity, Business model canvas and value proposition, start-up success and failure story, idea pitching (pitch deck development)

INTRODUCTION TO INTELLECTUAL PROPERTY RIGHTS (IPR)

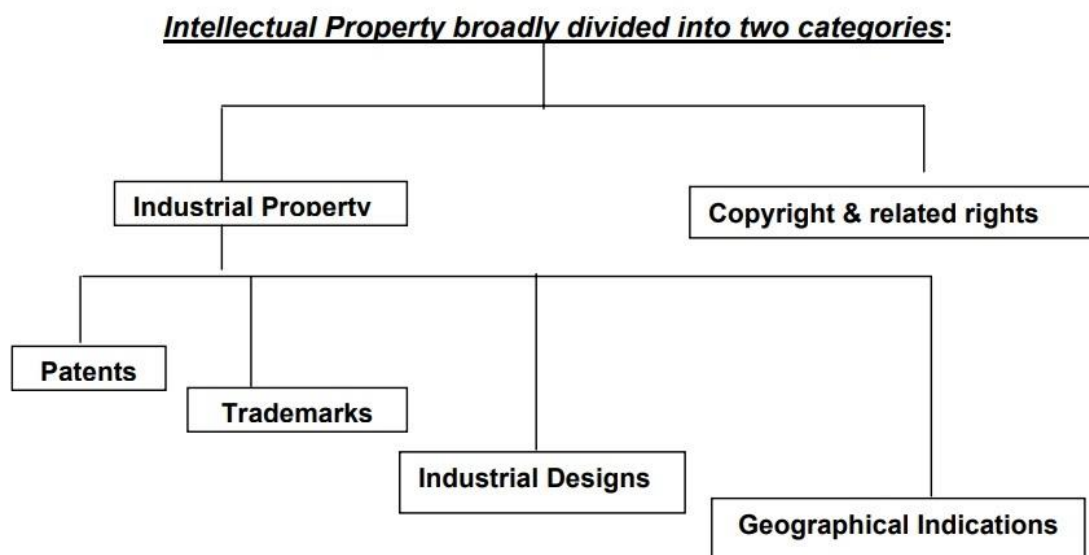
Intellectual Property (IP)

- It refers to creation of mind, such as inventions, literary and artistic works, designs, symbols, names and images which are protected by law.
- It allows creators or owners to control the use and exploitation of their creations for a certain period.
- It incentivizes innovation, creativity and economic growth by providing creator with the opportunity to benefit from their ideas and investments.

Intellectual property rights (IPR)

- A range of legal concepts aimed at protecting creations of the mind (e.g. inventions, designs, symbols, and images used in commerce).
- Key concepts include patents, copyrights, trademarks, trade secrets, and industrial design.
- These provides creators with exclusive rights to their respective creations, encouraging innovation and creativity while fostering economic growth.

Types of IPR



Here is a brief description of each type along with examples:

1. *Patents:*

- Patents grant inventors exclusive rights to their inventions
- These prevent others from making, using, selling or importing the invention without permission for a limited period (usually 20 years).
- Examples include pharmaceutical drugs, technological innovations like smartphones and industrial processes such as manufacturing methods.

2. *Copyrights*

- Copyrights protect original works of authorship, including literary, artistic, musical and dramatic works, as well as computer software and architectural designs.
- Copyright holders have the exclusive right to reproduce, distribute, perform and display their works.
- Examples include novels, paintings, songs, movies and software programs.

3. *Trademarks*

- Trademarks are signs, symbols, logos or names used to distinguish goods or services of one business from those of others.
- They provide exclusive rights to use the mark in commerce and prevent others from using confusingly similar marks.
- Examples include brand names like Coca-Cola, logos like the Nike swoosh, and slogans like "Just Do It".

4. *Trade secrets*

- Trade secrets are valuable, confidential information that gives a business a competitive advantage.
- They can include formulas, processes, designs, customer lists and marketing strategies.
- Unlike patents, trade secrets are not publicly disclosed, and their protection lasts as long as the information remains secrets and valuable.

- Examples include the Coca-Cola formula, the KFC recipe and Google's search algorithm.

5. *Geographical Indications (GIs)*

- Geographical indications identify products as originating from a specific geographical location and possessing qualities, reputation or characteristics attributable to that location.
- GIs protect traditional products and promote rural development by providing economic incentives to producers.
- Examples include Champagne (wine), Parmigiano-Reggiano (cheese) and Darjeeling (tea).

Advantages and Disadvantages of IPR

Advantages of IPR

1. *Encourages Innovation and Creativity*: IPR motivates inventors and creators to invest time, money, and effort in developing new products and ideas.
2. *Economic Growth and Competitive Advantage*: IPR can lead to increased investments and create jobs.
3. *Monetary Benefits*: Owners of IP can license or sell their rights, generating income.
4. *Brand Protection*: Trademarks protect brand identity, helping businesses distinguish their products and build consumer trust.
5. *Legal Protection Against Infringement*: IPR gives creators legal recourse if others use their work without permission, helping combat piracy and counterfeiting.
6. *Encourages Research and Development (R&D)*: Companies are more likely to invest in R&D when they can protect and profit from their innovations.

Disadvantages of IPR

1. *High Costs*: Registering and maintaining IP (especially patents) can be expensive, particularly for start-ups or small businesses.
2. *Complex Legal Processes*: Enforcing IPR can involve lengthy, complex, and costly legal battles.

3. *Limited Duration:* Most IP rights are time-limited (e.g., patents expire after 20 years), after which anyone can use the invention.
4. *Hindrance to Innovation:* Overprotection can stifle innovation, as others may be blocked from improving or building upon existing ideas.
5. *Access and Inequality Issues:* In sectors like healthcare, patent protections can limit access to life-saving medicines, particularly in low-income countries.
6. *Monopolistic Practices:* IP can lead to monopolies, where one company dominates the market, potentially leading to higher prices and limited choices.

PRIOR ART RESEARCH TECHNIQUES IN BUSINESS AND INNOVATION:

- In business and innovation, **prior art research** means investigating existing ideas, products, patents, research, and market solutions to check whether your new idea is **original, non-infringing, and commercially viable** and **not already protected or widely used**.
- It's a crucial step before launching a product, filing a patent, or pitching an idea.

Prior Art Research Techniques in Business and Innovations

1. Patent Database Search

- Look for similar inventions or solutions in patent records.
- Tools:
 - Google Patents
 - WIPO (World Intellectual Property Organization)
 - USPTO (United States Patent and Trademark Office)
 - Indian Patent Office database
- Helps identify legal risks and potential competitors.

2. Academic & Technical Literature Review

- Search journals, white papers, and technical reports to see if the concept has already been studied.

- Tools: Google Scholar, IEEE Xplore, ResearchGate, PubMed (for biotech/medical ideas).

3. Market Research

- Study existing businesses, start-ups, and product offerings in the market.
- Techniques: competitor analysis, product benchmarking, trade fairs, and industry reports (e.g., Gartner, Statista).

4. Trademark & Brand Check

- If your idea involves branding/names, check for registered trademarks.
- Tools:
 - USPTO TESS (for US)
 - EUIPO (for Europe)
 - IP India Public Search

5. Business & News Databases

- Use news archives, company reports, and start-up databases to spot existing players.
- Tools: Crunchbase, PitchBook, CB Insights, LinkedIn, Bloomberg.

6. Web & Open-Source Search

- General internet search (Google, Bing) to uncover blogs, forums, product launches, and concept discussions.
- Social media trend analysis (Twitter/X, LinkedIn, Reddit, Quora).

7. Industry Expert Consultation

- Interview professionals, mentors, and domain experts.
- Helps validate whether the idea is genuinely novel or if similar approaches already exist.

FREQUENT MYTHS ABOUT ENTREPRENEURS

1. Entrepreneurs don't have a personal life

Myth: Entrepreneurs should renounce their personal life.

Reality: Work-life balance is the key. The most successful entrepreneurs know how to take breaks and master their prioritizing, delegating, and scheduling skills.

2. Real entrepreneurs don't quit

Myth: It is probably of the most famous myths that real entrepreneurs never quit.

Reality: Business world is full of examples of people who have quit and benefited from it.

3. Business people don't have a boss

Myths: Business give you absolute independence.

Reality: Entrepreneurs, just like the rest of the world, need to establish connections if they want their business to thrive.

4. Entrepreneurs are rich

Myth: All that entrepreneur need is money.

Reality: It is true that a venture needs capital to survive. It is also true that a large number of business failures occur because of lack of adequate financing. However, money is not only bulwark against failure. Many successful entrepreneurs have overcome a "money problem" while establishing their ventures. To these entrepreneurs, money is a resource but never an end in itself.

5. The key to success is believing in it enough

Myth: Believing in your idea strongly makes it to the reality.

Reality: Believing in something will not automatically grant you success. Business requires perseverance, hard work, the capacity to understand your target and learn from your mistakes and willingness to innovate daily.

6. Entrepreneurs take extreme risks

Myth: Business people are addicted to taking risks.

Reality: The most successful people on the planet are very good at calculating risks.

7. Entrepreneurs think only about money

Myth: Entrepreneurs are in business for becoming rich.

Reality: They do it for different reasons, especially for pursuing their passion, obtaining financial independence, and having more flexibility.

8. There is no business without a unique idea (Entrepreneurs are always inventors)

Myth: The product or service should be unique for successful business.

Reality: There are thousands of businesses in the world with the same concept behind them, but it is the way they act that makes them different. Several businesses offer the same product, but they differ in marketing strategy, user experience and so on.

9. There is a secret to success

Myth: Only successful entrepreneurs know the secret to successful business.

Reality: There are no secrets, just the same old thing. Work hard on your idea and do not be afraid of failure. You have to learn from your mistakes and find a way to overcome them.

10. Entrepreneurs should always trust their guts

Myth: Successful entrepreneurs have strong intuition and they always trust their gut feeling.

Reality: There is nothing wrong to follow one's instinct but it is fundamental to listen to clients and value constructive feedback. There is always a way of improving and, sometimes, what we think of as genius ideas are not that brilliant.

11. You have to be young and restless to be an entrepreneur

Myth: An entrepreneur should be young to achieve success

Reality: There are sufficient exceptions to conclude that age matters, but not always: other things may matter more, such as the availability of start-up finance, outward and inward migration, social and cultural factors, alternative job opportunities or the lack of them, levels of human capital, and access to social security.

12. Entrepreneurs are born, not made

Myth: Entrepreneurial skills are innate, these can't be achieved by learning.

Reality: One can learn how to start business or become a leader.

13. It gets easier with time

Myth: Business gets easier with time.

Reality: A business never gets easier. You may enjoy it more over time. The issues may evolve from more tactical to more strategic as the business grows. If you do find it getting easier, you need to start finding challenges for yourself and your organization that will result in excellent growth opportunities.

14. Intelligence Quotient (IQ) is a primary indicator of entrepreneurial success

Myth: Success of an entrepreneur depends on IQ.

Reality: It's hard to downplay the value of being a smart business owner. However, there are two other qualities which are quite significant; *Emotional Quotient (EQ)* and *Adversity Quotient (AQ)*.

15. Entrepreneurs are individuals who work alone.

Myth: Entrepreneurs prefer to work alone.

Reality: Successful entrepreneurs find people with the right skills to join their team. Entrepreneurship is truly a team sport.

16. All that entrepreneurs need is luck.

Myth: One must be lucky to achieve success.

Reality: “*Being in the right place at right time*” is always an advantage but “*luck happens when preparation meets opportunity*” is an equally appropriate adage. Prepared entrepreneurs readily seize the opportunity when it arises and turn them into success.

BUSINESS MODEL CANVAS

Objective: To give a **clear, structured, and visual picture** of a business idea or model, making it easier to test, refine, and communicate.

- The **Business Model Canvas** is a strategic tool developed by **Alexander Osterwalder** and **Yves Pigneur**

- It helps entrepreneurs, start-ups, and organizations to **visualize, design, and analyze their business model** in a single page.
- It has **9 building blocks** given below:

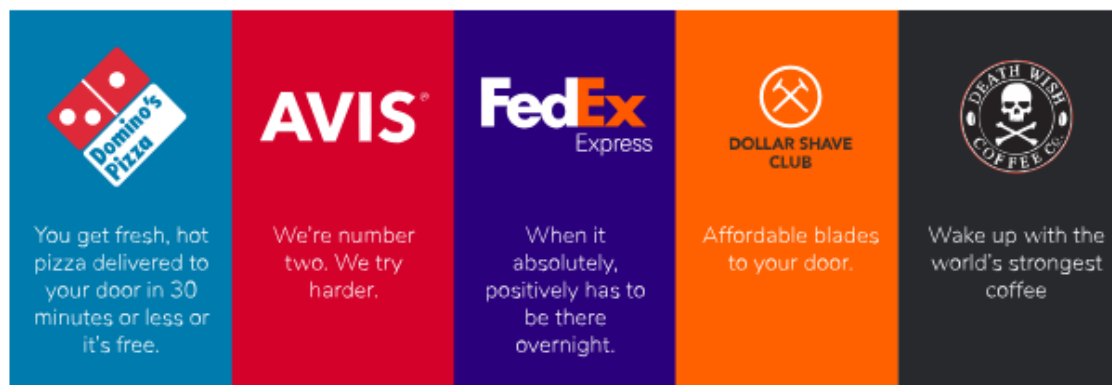
1. **Customer Segments** – Who are your target customers?
2. **Value Proposition** – What unique value are you offering them?
3. **Channels** – How do you deliver the value?
4. **Customer Relationships** – How do you interact and build loyalty?
5. **Revenue Streams** – How do you make money? (sales, subscriptions, ads, etc.)
6. **Key Resources** – What assets are essential (people, technology, IP, capital)?
7. **Key Activities** – What are the most important things you must do?
8. **Key Partners** – Who helps you (suppliers, collaborators, alliances)?
9. **Cost Structure** – What are the major costs to run the business?

Business Model Canvas		Designed for:	Designed by:	Date:	Version:	
Key Partners Complementary Core Competencies: Are there organizations that have specialties that we can leverage to support our business model and delivery of our value proposition to the customer? Resource Providers: Which Key Activities will be better performed by outside parties? Which Key Resources are you acquiring from partners?	Key Activities Core Competencies: What do you need to do to create / deliver this product or service? How will you and your team spend their time? What major operational tasks and processes do you have to excel in order to provide the value proposition in a repeatable and sustainable manner? Key Resources Required Assets: What are the most important assets required develop and maintain your core competency areas and make your business model work (physical, intellectual, human, financial, etc.)?	Value Propositions Gains (Value, Benefits, Outcomes): For each of the Jobs listed in Segments section, what does the customer want to accomplish? Functional Outcomes: What would the customer like to be able to do differently if they could? Social-Emotional Outcomes: How would the customer feel if they were able to do things differently (emotional value)? Pain Relief: What obstacles or pain points would the customer like to see diminished or eliminated? Initial Solution Concept: Does the solution under consideration have a unique value proposition?	Customer Relationships Engagement/Interaction: What type of experience does each customer segment/type expect at each touch point. Touch Points: How often and how do you 'touch' the customer and through what means? (i.e. social media, email, phone, advertising, personal interactions?) Channels Channel Selection: Corresponds directly with the customer touch points identified in Customer Relationship. The whole sales cycle from awareness to evaluation to purchase decision - delivery - post purchase service and engagement.	Customer Segments Customer Types/Context: Based on the similarity around job, context, and progress to be made, not demographics. Jobs: The underlying tasks that customers are trying to get done - What are they trying to accomplish? Under what context or circumstances? Pains: A pain point is an area where a customer experiences frustration, boredom, or inefficiency when trying to make		
Cost Structure What are the major cost items/expenses associated with your key activities and resource needs? Any major startup costs? Where do your major operational costs come from?		Revenue Streams What does each customer segment purchase? Are you selling a product? service? Do all customers buy the same thing or are they different? Are there more than one revenue stream? How much does each revenue stream contribute to overall gross profit?				

VALUE PROPOSITION

- Value propositions are arguably the most important element of the business model canvas template.

- “A value proposition is an end benefit a customer receives after purchasing your product, service or solution”.
- Generally, the value proposition is addressed to the company’s target customers or target market segment.
- *Essentially, a value proposition specifies what makes the company’s product or service attractive, why a customer should purchase it, and how the value of the product or service is differentiated from similar offerings.*
- The proposition takes the form of a short, clear, and concise statement of the tangible and intangible benefits that will be delivered to customers.
- The perfect proposition must quickly transmit the values to potential customers without the need for further explanation.



TYPES OF VALUE PROPOSITION

All products, services or solutions offer anyone or a combination of the following categories;

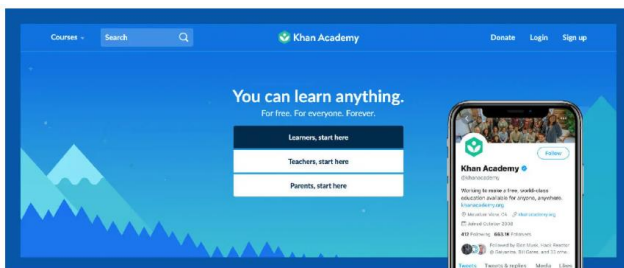
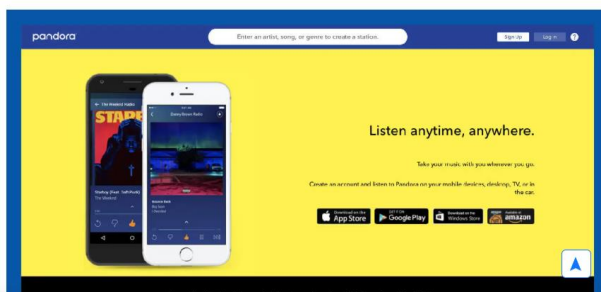
- ❖ **Functional value proposition:** These propositions emphasize how a product or service can make a customer's life easier, more efficient, or more productive.
- ❖ **Emotional value proposition:** While functional propositions appeal to logic, emotional value propositions target feelings and experiences.
- ❖ **Social value proposition:** Social value propositions appeal to customers' desire to belong to a group or to stand out from the crowd.
- ❖ **Cost value proposition:** Cost value propositions focus on saving customers money or time.

- ❖ **Innovation value propositions:** These propositions appeal to customers who are drawn to the latest technology or novel solutions.
- ❖ **Convenience value propositions:** These propositions emphasize simplicity, accessibility, and user-friendliness.

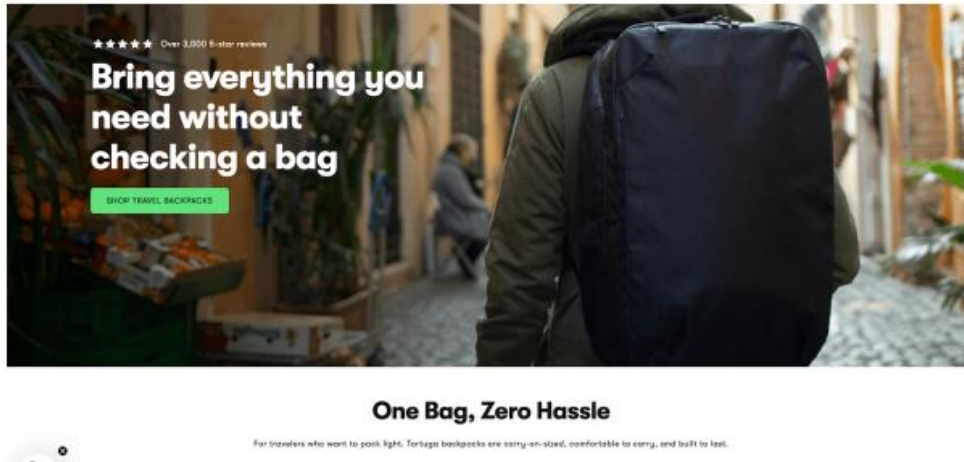
Examples of value proposition

Some examples of value propositions are listed below:

1. **Amazon Prime:** Fast, convenient, and unlimited free delivery.
2. **Pandora:** Listen anytime anywhere.
3. **Evernote:** Helps users tame their work and organize their life.
4. **Nike:** "Just do it," motivating and empowering customers.
5. **Coca-Cola:** Open Happiness, focusing on refreshment and enjoyment.
6. **Khan Academy:** You can learn anything.
7. **Tortuga Backpacks:** “Bring everything you need without checking a bag”, simple statement of how to use its backpacks.

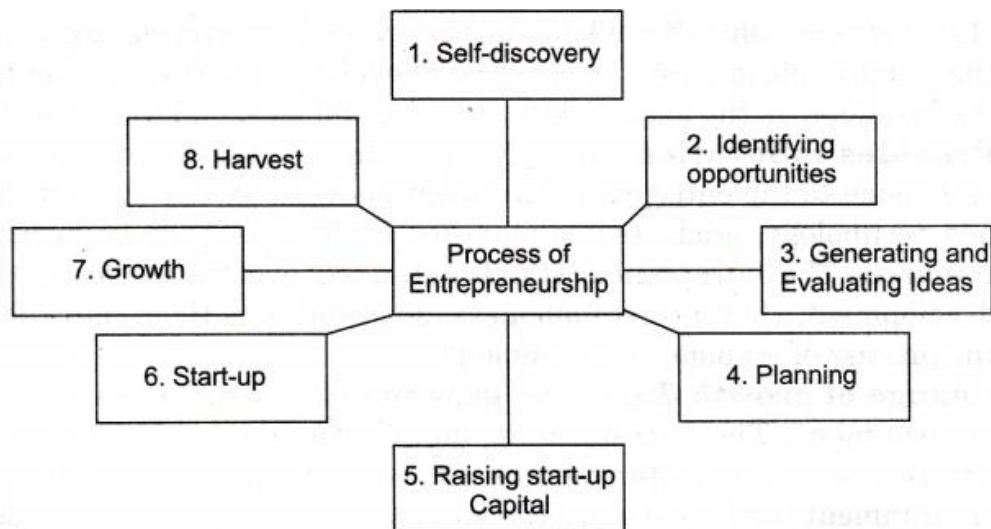


Tortuga Backpacks



CONCEPT OF ENTREPRENEURSHIP

- Entrepreneurship is the ability and readiness to develop, organize and run a business enterprise, along with any of its uncertainties in order to make a profit.
- The concept of entrepreneurship is the dynamic process of identifying opportunities, innovating, organizing resources, and undertaking risks to create and manage a new venture with the goal of generating value, profit, or social impact.
- It involves vision, innovation, calculated risk-taking, and the transformation of ideas into reality through the establishment and growth of new businesses, products, or services.



ENTREPRENEURSHIP COMPETENCIES

- Entrepreneurial competency is a set of skills and behaviour needed to create, develop, manage, and grow a business venture.
- It also includes the ability to handle the risks that come with running a business.
- Business owners and start-up founders must possess most of the entrepreneur competencies to succeed.
- Just like other types of competencies, these could be technical, behavioural, attitude-based, or productivity-based.

Major Entrepreneurial Competencies

Entrepreneurship Development Institute of India (EDII), Ahmedabad identified a set of entrepreneurial competencies or characteristics that results in superior performance.

1. Initiative: It is the entrepreneur who initiate a business activity.

2. Looking for opportunities: Entrepreneur looks for opportunity and takes appropriate actions as and when it arises.

3. Persistence: Entrepreneur follows the Japanese proverb “fall seven times; stand up eight”. Repeated efforts to overcome obstacles that get in the way of reaching goals.

- 4. Information seeker:** Takes individual research and consult experts to get information to help reach the goal.
- 5. Quality consciousness:** Entrepreneur takes way strong urge to excel to beat the existing standard.
- 6. Committed to work:** Does every sacrifice to get task completed.
- 7. Efficiency seeker:** Makes always tenacious efforts to get the task completed within costs and time.
- 8. Proper planning:** Formulates realistic and proper plans and then executes rigorously to accomplish the task.
- 9. Problem solver:** Always tries to find out ways and means to tide over the difficult times.
- 10. Self-confidence:** A strong believer in his strength and abilities.
- 11. Assertive:** Good in asserting his issues with others for the cause of employees
- 12. Persuasive:** Able to successfully convince others to do what he actually wants.
- 13. Efficient monitor:** Personally, supervises the work so that it is done as per the standards laid down.
- 14. Employee's well-wisher:** Has great concern and also takes necessary measures to improve the welfare of the employees working in the enterprise.
- 15. Effective strategy:** Introduces the most effective strategies to effect employees to achieve the enterprise goal whatsoever it may be.

SENSING BUSINESS OPPORTUNITIES

Opportunity Sensing: A good business opportunity is one that effectively meets a demand in the market with a unique solution and has the potential for growth and profitability.

Here are the key criteria to consider:

- *Market Demand:* Identify clear, existing or emerging demands through research and market analysis.
- *Unique Value Proposition:* Offer something distinct or superior to current market offerings.
- *Scalability:* Ensure the opportunity has potential for growth in products, services, or market reach.

- *Sustainability*: Aim for a long-term, sustainable model with consistent revenue generation and adaptability.
- *Profitability*: Have a clear path to profitability, with solid projections for revenue, costs, and margins.
- *Alignment with Skills and Interests*: Align the business with your passion and knowledge to increase perseverance and success.
- *Competitive Advantage*: Secure a competitive edge through technology, brand, knowledge, or customer relationships.
- *Market Trends*: Align with or stay ahead of market trends and technological advancements.
- *Regulatory Environment*: Understand and navigate the regulatory landscape, ensuring manageable compliance.
- *Social and Environmental Impact*: Consider the social or environmental impact, as sustainability and ethics are increasingly valued.

STARTUPS SUCCESS AND FAILURE STORIES

1. Bill Gates: Traf-O-Data hit fatal road bumps

- Bill Gates founded Traf-O-Data in 1972 wherein his system recorded traffic information and fed it back to government authorities, civil engineers and others.
- While it sounded like a useful business idea, the concept ultimately failed.
- Despite failure, the project gave Gates the experience and skills they needed to create Microsoft's first line of software products a few years later

Key Takeaway: Failed ventures provide valuable experience and skills that can be applied to future successful endeavours.

2. Steve Jobs: Fired by the company that he cofounded

- Apple I and Apple Lisa were two products produced by Apple that crashed and burned.
- It cost the company millions of dollars in development — which failed to recoup.

- A pattern of costly production decisions led to Apple's co-founder Steve Jobs's ouster from Apple in the mid-1980s.
- Jobs eventually found his way back to the company in 1997, through another business he founded in the interim, NeXT.
- After Apple acquired NeXT, Jobs himself became a valuable commodity again.

Key Takeaway: Being forced out of your own company doesn't end your story — it can create new opportunities for innovative comebacks.

Startups that tanked

1. Amazon Food, Distribution (Founded in 2020, Dissolved in 2022)

- In May 2020, Amazon Food (founder Jeff Bezos) entered the competitive Indian food delivery market.
- After trying it out for more than two and half years, Amazon decided to shut down its food delivery platform, being piloted in Bengaluru, India, by 29 December 2022.

Reasons for failure

- Stiff competition from established players like Zomato and Swiggy.
- Localization challenges in catering to diverse culinary preferences
- Operational complexities in building a reliable network of restaurants and delivery partners
- Broader cost-cutting measures undertaken by Amazon in a challenging economic environment.

IDEA PITCHING

Idea pitching

- It is the process of presenting a business idea, product, or solution to an audience (such as investors, potential partners, customers, or even internal teams) with the goal of gaining support, funding, or approval.

- It's essentially about **communicating your idea in a clear, concise, and convincing way** so that others see its value.

Key elements of idea pitching:

1. **Problem Statement** – What issue or gap are you addressing?
2. **Solution (Your Idea)** – How does your product/service/approach solve the problem?
3. **Value Proposition** – Why is your idea unique or better than existing solutions?
4. **Market Potential** – Who needs it? How big is the opportunity?
5. **Business Model** – How will it make money or sustain itself?
6. **Execution Plan** – How you'll implement it (resources, timeline, team).
7. **Call to Action** – What you want from your audience (funding, partnership, approval, etc.).

Purpose of idea pitching:

- To **attract investors** for funding.
- To **convince stakeholders** or management to support your idea.
- To **win competitions or grants**.
- To **build partnerships** and collaborations.

A good pitch is usually **short (2–10 minutes)**, uses **storytelling** to connect emotionally, and is backed by data or evidence to build credibility.

PITCH DECK DEVELOPMENT

Pitch Deck:

- A pitch deck is often used to raise capital, attract talent, and build partnerships.
- A pitch deck is a visual presentation that summarizes your start-up's vision, mission, and goals.
- It is typically a 10-15 slide presentation that provides an overview of your business, highlighting its unique value proposition, market opportunity, competitive advantage, and growth potential.

Significance of Pitch Deck: Pitch decks are essential for start-up founders for several reasons:

1. *Attracts Potential Investors*
2. *Communicates the Business Model Clearly*
3. *Establishes Credibility*
4. *Networking*
5. *Facilitates Fundraising*

Key Elements of a Successful Pitch Deck

A successful pitch deck should include the following 6 key elements:

1. *Problem Statement:* Clearly define the problem your start-up solves, and the market needs it addresses.
2. *Product or Service Overview:* Provide a concise yet comprehensive explanation of the product or service, its unique features, and its competitive advantages.
3. *Market Opportunity:* Outline the size and growth potential of the target market, as well as the competitive landscape.
4. *Business Model and Revenue Strategy:* Outline the business model, including the revenue streams, pricing structure, and scalability potential.
5. *Team and Advisors:* Introduce the founding team, highlighting their relevant experience, expertise, and track record, as well as any advisors or strategic partners involved.
6. *Financial Projections:* Include realistic financial projections, such as revenue forecasts, expenses, and key metrics, demonstrating the potential for growth and profitability.